Grain Definition Document — Courier & Logistics DW

Student Name / ID: Taha Imran /24L-2560, Muhammad Ibrahim/24L-2602, Muhammad Ayan /24L-2568, Rai Fahd/ 24L-2509
Project Title: Enterprise Courier & Logistics Data Warehouse Solution (LogiTrack DWBI)
Module: DL3004 Data Warehousing & Business Intelligence
Phase: Phase 1 (30% Weightage)
Methodology: Ralph Kimball Dimensional Modeling Lifecycle

1. Executive Summary

In the **Kimball Dimensional Modeling Lifecycle**, declaring the grain is the second and most critical step of the four-step dimensional design process (Step 1: Select Business Process; **Step 2: Declare Grain**; Step 3: Identify Dimensions; Step 4: Identify Facts).

According to Ralph Kimball:

> *"The grain is the operational definition of what a single fact table record represents. You cannot identify dimensions or facts until you have formally declared the grain. Violating grain consistency corrupts aggregations, causes Cartesian products, and destroys analytical credibility."*

This document formalizes the **precise grain declarations** for all three fact tables in the LogiTrack data warehouse, providing operational generation triggers, atomic granularity justifications, and key uniqueness definitions.

2. Master Grain Declarations Summary

Fact Table Name	Fact Archetype	Formal Grain Statement
`Fact_Delivery_Attempt`	Transaction Fact	"One row represents a single physical delivery attempt event made by a field rider for an individual parcel package at a specific recipient address at a specific point in time."
`Fact_Daily_Hub_Backlog`	Periodic Snapshot Fact	"One row represents the cumulative inventory count and dwell hours for a specific parcel type in a specific backlog status at a specific logistics hub at the end of each operating day (23:59:59)."

`Fact_Rider_Route_Assignment`	**Factless Fact**	**"One row represents the scheduled workforce deployment assignment of an individual rider to a designated delivery route corridor, operating hub, and working shift for a specific calendar date."**

3. Deep-Dive Grain Analysis by Fact Table

3.1 Fact Table 1: Fact_Delivery_Attempt (Transaction Fact)

Fact_Delivery_Attempt — Grain Specification

> *"One row represents a single physical delivery attempt event made by a field rider for an individual parcel package at a specific recipient address at a specific point in time."*

Operational Trigger & Frequency: Triggered whenever a rider scans a parcel at the delivery doorstep using the Mobile Dispatch App, logging an outcome (Delivered with signature/photo, or a logged exception such as "Customer Unavailable" or "Gate Locked"). Ingested hourly via micro-batch delta scraping.

Key Formulations: Surrogate Primary Key: `Delivery_Attempt_Key` (INT) | Natural Business Key: `Attempt_ID` (UUID) | Composite Grain Uniqueness: `Tracking_Number` + `Attempt_Sequence_Number`.

Grain Justification (Atomic Attempt vs. Coarse Shipment): Setting the grain to "one row per shipment" would force overwriting previous attempts or creating flat columns (`Attempt_1_Status`, `Attempt_2_Status`), which destroys failure root-cause analysis and conceals the marginal operating costs of re-deliveries. The atomic attempt grain preserves every event independently, making it trivial to compute **First-Attempt Delivery Rate (FADR)** via `WHERE Attempt_Sequence_Number = 1`.

Dimensional Intersection: Intersects with Date (Attempt & SLA Deadline), Origin Hub, Route, Rider, Customer, Service Type, Delivery Status (Junk Dimension), and Parcel Type.

Measure Validity: Fully additive facts (`Delivery_Fee_Amount`, `Estimated_Labor_Cost`, `Estimated_Fuel_Cost`, `Net_Margin_Amount`). *Rule:* `Delivery_Fee_Amount` is billed once on `Attempt_Sequence_Number = 1`, while secondary re-attempts record $0.00 revenue and capture strictly marginal labor and fuel expenses.

3.2 Fact Table 2: Fact_Daily_Hub_Backlog (Periodic Snapshot Fact)

Fact_Daily_Hub_Backlog — Grain Specification

> *"One row represents the cumulative inventory count and dwell hours for a specific parcel type in a specific backlog status at a specific logistics hub at the end of each operating day (23:59:59)."*

Operational Trigger & Frequency: Automated batch process executing at 23:59:59 midnight following evening sorting and outbound line-haul dispatches. Ingested daily at 00:30 AM.

Key Formulations: Surrogate Primary Key: `Hub_Backlog_Key` (INT) | Composite Natural Business Key: `Snapshot_Date` + `Hub_Code` + `Parcel_Type_Code` + `Backlog_Status`.

Grain Justification (Periodic Snapshot vs. Event Scans): Calculating historical backlog by subtracting cumulative outbound scans from inbound scans across 10 million raw barcode events takes 15–30 seconds per query. The snapshot grain pre-aggregates and freezes the inventory state into a compact table of only ~100 rows per day, enabling sub-second 12-month trend queries.

Dimensional Intersection: Intersects strictly with Date (Snapshot Date), Hub, and Parcel Type. Route, Rider, and Customer are absent because staged backlog is an internal facility state prior to route dispatch.

Measure Validity: Semi-additive measures (`Backlog_Parcel_Count`, `Average_Dwell_Hours`, `Max_Dwell_Hours`) sum across hubs for a single date, but must be averaged across time. Fully additive measures: `Daily_Inflow_Count`, `Daily_Outflow_Count`.

3.3 Fact Table 3: Fact_Rider_Route_Assignment (Factless Fact)

Fact_Rider_Route_Assignment — Grain Specification

> *"One row represents the scheduled workforce deployment assignment of an individual rider to a designated delivery route corridor, operating hub, and working shift for a specific calendar date."*

Operational Trigger & Frequency: Triggered when the dispatch supervisor locks the morning shift roster in the Workforce Management System at 06:00 AM daily.

Key Formulations: Surrogate Primary Key: `Assignment_Key` (INT) | Composite Natural Business Key: `Assignment_Date` + `Rider_ID` + `Route_Code` + `Shift_Code`.

Grain Justification (Pure Factless vs. Mixed Metrics): Storing delivery execution metrics (delivered counts, hours worked) in the assignment table violates Kimball's single-grain rule by mixing planning (06:00 AM) with execution (all-day). Storing hours worked in the table violates the factless fact pattern. Keeping it pure keys allows **non-event tracking** (outer-joining to find unassigned routes). **Deliveries per Rider-Hour (DPRH)** is computed cleanly via multi-fact drill-across using standard hours from `Dim_Shift`.

Dimensional Intersection: Intersects with Date (Assignment Date), Rider, Route, Hub, and Shift.

Measure Validity: Contains **zero numeric measure columns**. Supports `COUNT(Assignment_Key)` for total scheduled shifts.

4. Comprehensive Fact Grain Comparison Matrix

Architectural Property	Fact_Delivery_Attempt	Fact_Daily_Hub_Backlog	Fact_Rider_Route_Assignment

Fact Archetype	Transaction Fact	Periodic Snapshot Fact	Factless Fact (Coverage)
Business Process	Last-Mile Delivery Attempts	Hub Sorting Inventory Balances	Workforce Route Scheduling
Formal Grain	1 row per physical delivery attempt for an individual parcel.	1 row per hub, date, parcel type, and backlog status at 23:59:59 daily.	1 row per scheduled rider assignment to route, hub, shift, date.
Temporal Cadence	Event-driven (hourly micro-batch).	Daily snapshot (frozen at 23:59:59).	Daily planning batch (06:00 AM).
Surrogate Primary Key	`Delivery_Attempt_Key` (INT)	`Hub_Backlog_Key` (INT)	`Assignment_Key` (INT)
Natural Composite Key	`Tracking_Number` + `Attempt_Sequence_Number`	`Snapshot_Date` + `Hub_Code` + `Parcel_Type_Code` + `Status`	`Assignment_Date` + `Rider_ID` + `Route_Code` + `Shift_Code`
Daily Record Volume	15,000 – 25,000 rows / day	60 – 120 rows / day	200 – 400 rows / day
Annual Table Growth	~5.5 to 9.1 million rows / year	~25,000 to 45,000 rows / year	~75,000 to 150,000 rows / year
Measure Types	Fully Additive (Fee, Cost, Margin, Duration (Weight is non-additive across repeat attempts)).	Semi-Additive (Backlog Count, Dwell Hours); Fully Additive (Inflow/Outflow).	**None** (Foreign keys only; zero numeric measures).
Primary KPI	On-Time Delivery Rate (OTDR), First-Attempt Delivery Rate (FADR), Route Margin.	Hub Backlog Ratio (HBR %), Average Parcel Dwell Time.	Route Schedule Coverage Ratio, Deliveries per Rider-Hour (via drill-across).